

AN INTERNSHIP REPORT
“INTERNSHIP REPORT ON CIVIL ENGINEERING WORKS
AT TARAKESHWOR MUNICIPALITY”

Submitted in partial fulfilment of the requirements for

BACHELOR’S DEGREE

in

CIVIL ENGINEERING

by

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Internship carried out at:

Department of Infrastructure,
Tarakeshwor Municipality, Municipal Executive Office

UNDER THE GUIDANCE OF

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External Supervisor: Er. Rakesh Sunari Magar, Ast. Executive Engineer



NEPAL COLLEGE OF INFORMATION TECHNOLOGY

(AFFILIATED TO POKHARA UNIVERSITY)

BALKUMARI, LALITPUR

24th August, 2026



“पढिपल, अपनत्व र अधिकार : बिगो विकासको आधार”
तारकेश्वर नगरपालिका
तानकान्न नगरपालिका
TARAKESHWOR MUNICIPALITY
OFFICE OF THE MUNICIPAL EXECUTIVE
धर्मस्थली, काठमाडौं, बागमती प्रदेश, नेपाल
DHARMASTHALI, KATHMANDU, BAGAMATI PROVINCE, NEPAL



पत्र संख्या (Ref. No.): ०८३/०८४

चलानी नं. (Dispatch No.): ४८४



मिति (Date): २०८३/५/११

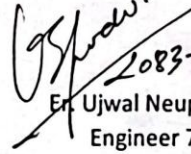
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CERTIFICATE OF INTERNSHIP COMPLETION

This is to certify that Binod Raj Regmi, a student of the Bachelor of Civil Engineering Program at Nepal College of Information Technology (NCIT) has successfully completed an internship at Tarakeshwor Municipality, Dharmasthali, Kathmandu from Baishak 10, 2083 BS to Shrawan 10, 2083 BS.

During this Internship, Binod Raj Regmi demonstrated exceptional dedication, professional skills, and the ability to apply theoretical knowledge to practical challenges. The internship provided significant exposure to Site Supervision, Site Measurement, Rate Analysis and Estimation. Also, the intern verified the cadastral map in the field and assisted in office work such as preparation and verification of measurement books, final bills, material breakdowns and related project documentation.

We commend Binod Raj Regmi for his achievements and wish him continued success in his future endeavors.


2083-5-11
En. Ujwal Neupane
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Tarakeshwor Municipality
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This is to certify that Binod Raj Regmi has completed the internship at Tarakeshwor Municipality from Baishakh 10 to Shrawan 10, 2083 B.S., as a part of the requirements for the Bachelor's Degree in Civil Engineering at Nepal College of Information Technology (NCIT). The internship activities and this report were monitored and guided from the college side under my supervision.

This internship report is hereby recommended and forwarded for evaluation.

.....

Er. Narayan Prasad Regmi

Department of Civil Engineering,

Nepal College of Information Technology (NCIT)



NEPAL COLLEGE OF INFORMATION TECHNOLOGY

(AFFILIATED TO POKHARA UNIVERSITY)

Issued on: 26 August 2026

CERTIFICATE OF INTERNSHIP COMPLETION

This is to certify that **Mr. Binod Raj Regmi**, a student of the **BE Civil Engineering** at **Nepal College of Information Technology (NCIT)**, has successfully completed an internship at **Tarakeshwar Municipality** from **23rd April 2026** to **14th August 2026**.

During this internship, **Binod Raj Regmi** demonstrated exceptional dedication, professional skills, and the ability to apply theoretical knowledge to practical challenges. The internship provided significant exposure to **Site Supervision, Site Measurement, Rate Analysis and Estimation**. Also, the intern verified the cadastral map in the field and assisted in office work such as **preparation and verification of measurement books, final bills, material breakdowns and related project documentation**.

We commend **Binod Raj Regmi** for his achievements and wish him continued success in his future endeavors.

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ACKNOWLEDGEMENT

With great pride, I would like to express my gratitude to Nepal College of Information Technology and Tarakeshwor Municipality for providing me the required platform for the fulfillment of the internship.

I express wholehearted gratitude to dear sir Mr. Narayan Puri, respected Head of Civil Engineering Department. I wish to acknowledge his support for making my task easy by providing me with all valuable help and encouragement.

I am extremely grateful to my supervisor Mr. Rakesh S Magar, Assistant Executive Engineer, for his guidance and support provided during the internship.

I would also like to thank all my senior colleagues, especially Ms. Muna Neupane and Nil Raj Vetuwal Sir who worked along with me in Tarakeshwor Municipality and shared their valuable knowledge and experience.

Finally, I would like to thank my family members and friends for their kind co-operation and motivation to proceed in my internship work.

Binod Raj Regmi (221808)

ABSTRACT

The internship training provided by the Infrastructure Department of Tarakeshwor Municipality was a valuable component of my fourth-year undergraduate study in Civil Engineering. It provided an opportunity to connect the theoretical knowledge acquired through academic coursework with its practical application in real-world engineering projects. During the internship, I observed construction practices, learned about municipal infrastructure development procedures, and developed a better understanding of the responsibilities of a civil engineer. The experience also improved my technical skills, professional confidence, analytical ability, and understanding of construction activities under actual field conditions.

I completed my internship at Tarakeshwor Municipality from 23 April to 23 July 2026. During this period, I was exposed to various civil engineering activities related to municipal infrastructure development. The major activities included project estimation, valuation of construction works, assessment of construction material quality, and observation of safety practices at project sites. These activities helped me understand the methods used for quantity calculation, project cost estimation, and evaluation of construction materials and workmanship. I also learned the importance of maintaining appropriate safety measures to reduce risks and promote a safe working environment.

The internship also provided insight into the coordination and collaboration required among the professionals and stakeholders involved in construction projects. I observed interactions among architects, civil engineers, technicians, contractors, labourers, and other stakeholders during project activities. This experience demonstrated that successful project completion depends not only on technical knowledge but also on effective communication, teamwork, planning, coordination, and professional responsibility. Working in a municipal environment further helped me understand the role of local government in infrastructure development and the importance of following approved procedures, specifications, and safety requirements.

Overall, the internship significantly enhanced my practical knowledge and professional skills. It enabled me to relate classroom theories to field practices and develop a better understanding of the challenges associated with construction and infrastructure development. The knowledge and experience gained during the internship will support my future academic work and professional career as a civil engineer.

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LIST OF ABBREVIATIONS

AutoCAD	Automatic Computer-Aided Design
BE Civil	Bachelor of Civil Engineering
BOQ	Bill of Quantities
c/c	Centre to Centre
DUDBC	Department of Urban Development and Building Construction
EPS	Expanded Polystyrene
KML	Keyhole Markup Language
METWET	Metal Weight Calculation Application
M20	Grade of Concrete with a Characteristic Compressive Strength of 20 MPa
MS	Mild Steel
NCIT	Nepal College of Information Technology
RCC	Reinforced Cement Concrete
SS 2402	Standard Specification Reference for Gabion Works
VAT	Value Added Tax

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Civil engineering is a discipline that integrates technical knowledge, practical application, and professional judgment in the planning, design, construction, and maintenance of infrastructure. Municipal engineering is an important part of this field because it supports local development through roads, public facilities, slope protection works, and other related services. The internship at Tarakeshwor Municipality provided an opportunity to understand how civil engineering functions in the context of local government administration and service delivery.

This report documents the internship completed in the Infrastructure Department of Tarakeshwor Municipality from 23 April 2026 to 23 July 2026. The internship was conducted as a required academic component of the Bachelor of Civil Engineering program at Nepal College of Information Technology (NCIT). It focused on understanding municipal engineering functions, reviewing construction-related documents, observing field works, and learning the process of estimation and rate analysis. Through this internship, I gained valuable exposure to both office-based and field-based practice in municipal civil engineering.

1.2 Background of the Works

Tarakeshwor Municipality is a local government body in Kathmandu District, responsible for local governance and infrastructure-related development within its administrative area. The municipality undertakes projects and works associated with road development, public buildings, slope protection, drainage, maintenance, and general service improvement. The Infrastructure Department plays a central role in the technical and administrative management of such works.

Municipal engineering activities require coordination among administrative staff, engineers, field technicians, surveyors, and local stakeholders. The internship therefore provided a practical opportunity to observe how engineering works are planned, reviewed, monitored, and implemented in a municipal setting. The work included review of municipal documents, quantity estimation, rate analysis, and field inspection of ongoing construction works.

1.3 Objectives

The main objective of the internship was to gain practical knowledge and field experience in civil engineering by understanding the application of theoretical concepts in actual municipal infrastructure projects. The internship was carried out at the Infrastructure

Department of Tarakeshwor Municipality, providing exposure to both office-based engineering activities and field-level construction works.

The specific objectives of the internship were:

- To understand the role, functions, and working procedures of the Infrastructure Department of a local government.
- To observe and gain practical experience in site surveying, measurement, and setting out of construction works.
- To understand the preparation and use of project proposals, BOQs, cost estimates, and rate analysis.
- To gain knowledge of quantity calculation, measurement, and valuation of construction works.
- To observe construction activities and understand the principles of construction supervision and quality control.
- To identify and understand the use and assessment of common construction materials in municipal projects.
- To gain practical exposure to road, drainage, retaining wall, gabion wall, and building construction works.
- To understand the application of engineering drawings, municipal maps, and technical documents in construction and land-related activities.
- To observe the coordination between engineers, surveyors, contractors, local residents, and other stakeholders during project implementation.
- To develop professional skills such as communication, teamwork, technical documentation, problem-solving, and site safety awareness.
- To relate the theoretical knowledge gained during academic study to actual civil engineering practices.
- To develop a better understanding of the responsibilities and professional role of a civil engineer in municipal infrastructure development.

1.4 Scope and Limitation

The internship covered a practical range of municipal civil engineering activities, while its limitations were mainly due to the nature of an internship role and the administrative constraints of a public office.

1.4.1 Scope

- Observation of municipal engineering works and construction procedures.
- Review of municipal construction documents, project proposals, and technical records.
- Participation in project estimation and rate analysis.

- Field visits to ongoing construction sites.
- Exposure to civil engineering practice in a local government setting.

1.4.2 Limitations

- Limited internship duration of three months
- Inability to observe every project stage
- Supervised rather than independent involvement
- Limited access to confidential project information
- Limited access to detailed structural calculations
- Difficulty locating some items in technical documents
- Limited availability of current market rates
- Limited use of advanced surveying instruments
- Limited exposure to laboratory testing procedures
- Limited experience with structural design software
- Dependence on ongoing project schedules
- Effects of weather and site accessibility
- Observation limited to assigned projects and activities

CHAPTER 2: ORGANIZATION DETAILS AND LITERATURE REVIEW

2.1 Introduction to Organization



Figure 1 Tarakeshwar Municipality Logo



Figure 2 Tarakeshwar Municipality Office

Tarakeshwar Municipality is a local government body located in Kathmandu District of Bagmati Province, Nepal. It was established in 2014 AD through the restructuring of existing Village Development Committees in accordance with the administrative restructuring of local governments in Nepal. The municipality lies in the northern part of Kathmandu Valley and covers a developing urban and semi-urban area with increasing residential, commercial, and infrastructural activities.

The municipality is administratively divided into 11 wards, each serving as a local administrative unit. The municipal office is responsible for delivering various public services and implementing development programs within its jurisdiction. Its major responsibilities include local infrastructure development, road construction and maintenance, drainage and water supply, building regulation, land-use management, public facilities, environmental management, and other services assigned to local governments under prevailing laws and regulations.

Tarakeshwar Municipality has a diverse physical setting consisting of relatively developed urban areas as well as sloping and semi-rural areas. The variation in terrain creates different requirements for infrastructure development, including road improvement, retaining structures, drainage systems, public buildings, and slope protection works. Rapid urbanization and population growth have further increased the demand for planned and sustainable infrastructure within the municipality.

2.2 Organizational Hierarchy

Tarakeshwar Municipality follows a structured organizational hierarchy to ensure effective administration, service delivery, and implementation of development activities. The municipality operates under an elected local government system, with the Municipal

Assembly as the main policy-making body. The mayor serves as the head of the municipal executive, while the Deputy Mayor assists in municipal administration and carries out responsibilities assigned under the prevailing laws and regulations.

Under the municipal executive, the Chief Administrative Officer (CAO) is responsible for the overall administrative management and coordination of municipal activities. Various branches and departments operate under the municipal administration to handle specific functions such as administration, finance, social development, education, health, planning, and infrastructure development.

The Infrastructure Department is responsible for technical activities related to the planning, development, construction, supervision, and maintenance of municipal infrastructure. The department is supported by engineers, sub-engineers, surveyors, and other technical personnel.

2.3 Working Domains of Organization

Tarakeshwor Municipality works across various sectors to provide public services and promote the overall social, economic, and physical development of the municipality. As a local government institution, its activities cover administrative services, infrastructure development, social development, environmental management, and other functions assigned to local governments. The major working domains of the municipality include:

- **Infrastructure Development:** Planning, construction, improvement, and maintenance of local roads, drainage systems, retaining structures, public buildings, and other municipal infrastructure.
- **Road and Transportation:** Construction and upgrading of local roads, road widening, maintenance, improvement of roadside drainage, and development of transportation facilities.
- **Building and Urban Development:** Building permit processing, implementation of building standards, building inspection, settlement development, and management of urban infrastructure.
- **Water Supply and Sanitation:** Coordination and implementation of water supply, drainage, sanitation, and waste-management-related activities.
- **Land and Survey-Related Services:** Verification of land boundaries, map-related works, site verification, land-use matters, and coordination with survey and land administration authorities.
- **Project Planning and Implementation:** Identification of development needs, preparation of project proposals, cost estimation, budgeting, implementation, monitoring, measurement, valuation, and completion of municipal projects.
- **Social Development and Public Services:** Provision and coordination of services related to education, health, social welfare, community development, and public facilities.
- **Environmental Management:** Activities related to environmental protection, solid waste management, drainage, greenery, and sustainable development.

- **Administrative and Financial Management:** Management of municipal records, revenue collection, budgeting, procurement, documentation, and administrative services.
- **Disaster Risk Management:** Activities related to disaster preparedness, risk reduction, slope stabilization, emergency response, and protection of vulnerable infrastructure.

Among these domains, the Infrastructure Department is particularly relevant to civil engineering. It deals with technical aspects of municipal development, including surveying, site measurement, preparation of estimates and BOQs, rate analysis, construction supervision, quality assessment, project valuation, and technical documentation. During the internship, practical exposure was mainly obtained in these engineering-related activities through both office work and field visits.

2.4 Description of Intern Department or Unit

The internship was carried out in the Infrastructure Department of Tarakeshwor Municipality, which is responsible for the technical planning, development, implementation, supervision, and maintenance of municipal infrastructure projects. The department plays an important role in translating the municipality's development plans into practical engineering works within its jurisdiction.

The department is involved in various activities related to road construction and maintenance, drainage works, retaining and gabion structures, public buildings, and other local infrastructure projects. It also performs technical and administrative tasks required for project implementation, including site inspection, surveying and measurement, preparation of project proposals, quantity calculation, cost estimation, preparation and review of Bills of Materials (BOMs), rate analysis, construction supervision, valuation, and project documentation.

The department consists of technical personnel such as engineers, sub-engineers, surveyors, and other supporting staff, who work together to carry out engineering activities. Engineers are generally responsible for technical planning, design review, estimation, supervision, and overall technical coordination, while sub-engineers and surveyors assist with field measurements, site verification, setting out, and other technical tasks. The department also coordinates with contractors, consultants, local residents, and other municipal sections during project implementation.

During the internship, I worked under the Infrastructure Department under the supervision of Assistant Engineer Rakesh S. Magar. I was exposed to both office-based and field-based activities. In the office, I reviewed engineering documents such as project proposals, BOQs, estimates, valuation documents, drawings, and other technical records. In the field, I participated in site visits involving site measurement, boundary verification, setting out, construction observation, material assessment, and supervision of ongoing works.

The department provided a practical environment for understanding the complete process of municipal infrastructure development, from project identification and estimation to implementation, measurement, valuation, and documentation. The experience helped bridge the gap between theoretical knowledge acquired during academic study and its practical application in civil engineering projects.

2.5 Literature Review / Related Study

The internship involved practical exposure to various civil engineering activities carried out by the Infrastructure Department of Tarakeshwor Municipality. To understand these activities effectively, relevant concepts, engineering standards, guidelines, and construction practices were reviewed. The study focused mainly on municipal infrastructure development, surveying, quantity estimation, construction materials, structural works, road and drainage works, and construction supervision.

2.5.1 Municipal Infrastructure Development

Municipal infrastructure includes the physical facilities and structures required to support the daily activities and development of a community. Local infrastructure works generally include roads, drainage systems, retaining structures, public buildings, water supply facilities, and other public facilities. Proper planning, estimation, implementation, supervision, and maintenance are essential for ensuring the functionality, safety, durability, and cost-effectiveness of such infrastructure.

2.5.2 Surveying and Site Measurement

Surveying is an essential part of civil engineering used to determine the location, dimensions, levels, and boundaries of land and construction sites. Site measurements and boundary verification are carried out before construction to ensure that the proposed works are located correctly. Cadastral maps and other survey records are also used for verifying land boundaries and identifying the relationship between the proposed project and existing land parcels. Accurate measurement and setting out help minimize errors during construction.

2.5.3 Quantity Estimation and BOQ

Quantity estimation is the process of determining the quantities of different materials and construction activities required for a project. It forms an important basis for determining the overall project cost. A **Bill of Quantities (BOQ)** provides a systematic listing of construction items along with their respective quantities and units. Estimation generally involves studying drawings, taking measurements, calculating quantities, applying appropriate rates, and determining the total estimated cost. Accurate estimation helps in budgeting, procurement, tendering, and financial management of construction projects.

2.5.4 Rate Analysis and Valuation

Rate analysis is used to determine the unit cost of a particular construction activity by considering materials, labour, equipment, transportation, overheads, and other applicable costs. The calculated rates are used in preparing project estimates and BOQs. During

project implementation, valuation is carried out based on the actual quantities of work completed and the applicable rates. Proper measurement and valuation are important for ensuring transparent and accurate payment to contractors.

2.5.5 Construction Materials and Quality Control

The quality of construction materials directly affects the strength, durability, and performance of infrastructure. Common materials used in municipal construction include cement, fine aggregate, coarse aggregate, reinforcement steel, masonry units, concrete, and stone. Quality control involves checking the suitability and required properties of materials and ensuring that construction activities follow the specified technical requirements. For concrete works, appropriate practices related to batching, mixing, placing, compaction, curing, and testing are necessary to achieve the required strength and durability.

2.5.6 Building and RCC Construction

Reinforced Cement Concrete (RCC) is widely used in building construction because concrete provides good compressive strength while reinforcement provides tensile strength. RCC construction requires proper placement and tying of reinforcement, formwork, concrete placement, compaction, and curing. Construction activities must follow approved structural drawings and specifications. Proper supervision is necessary to ensure that dimensions, reinforcement details, concrete quality, and workmanship meet the required standards.

2.5.7 Retaining and Gabion Structures

Retaining structures are constructed to retain soil and provide stability to slopes, roadsides, and excavated areas. RCC retaining walls and gabion walls are commonly used depending on site conditions, structural requirements, and available materials. Gabion walls consist of stone-filled wire-mesh baskets and are particularly useful for erosion control, slope stabilization, and protection of roadsides. Proper foundation preparation, alignment, drainage, stone placement, and installation of the wire baskets are important for their performance.

2.5.8 Road and Drainage Works

Road construction involves several activities including site preparation, earthwork, subgrade preparation, construction of pavement layers, drainage, and finishing works. Proper drainage is essential for protecting roads from water-related deterioration and maintaining their serviceability. Side drains and other drainage structures are provided to collect and safely convey surface runoff away from the road and surrounding areas. The design and construction of these facilities depend on site conditions, terrain, rainfall, traffic requirements, and applicable engineering standards.

2.5.9 Construction Supervision and Safety

Construction supervision is an important responsibility of technical personnel to ensure that works are executed according to approved drawings, specifications, quantities, and quality

requirements. It involves regular site inspection, measurement of completed works, monitoring of materials and workmanship, identification of construction issues, and coordination with contractors and stakeholders. Construction safety is also essential to protect workers, site visitors, and surrounding property. Appropriate personal protective equipment, safe working practices, proper site organization, and awareness of construction hazards are important aspects of safe construction.

2.5.10 Relevant Codes, Standards, and Guidelines

The planning and execution of civil engineering works require compliance with applicable national standards, codes, specifications, and municipal requirements. Relevant documents may include the **Nepal Building Code (NBC)** for building and structural works, applicable government specifications and norms for estimation and construction, and guidelines issued by relevant authorities for roads and other infrastructure. These standards provide technical requirements and recommended practices related to structural safety, materials, construction, measurement, and quality control.

The above theoretical concepts provided the foundation for understanding the practical activities observed during the internship. The knowledge gained through academic study and related references was compared with actual municipal practices during site visits, document review, measurement, estimation, and construction supervision activities.

CHAPTER 3: INTERNSHIP ACTIVITIES

The internship at Tarakeshwor Municipality provided me with valuable exposure to the practical and professional aspects of civil engineering. Through my involvement in municipal documentation, field visits, site surveying, construction observation, rate analysis, and project estimation, I was able to relate the theoretical knowledge gained during my BE Civil studies to actual engineering activities.

The training helped me develop technical understanding, practical skills, professional confidence, and awareness of the responsibilities of a civil engineer. It also provided opportunities to observe how engineers and technical personnel assess site conditions, prepare engineering documents, evaluate construction costs, monitor ongoing works, and coordinate with different stakeholders.

In addition to technical learning, the internship improved my communication, teamwork, time management, problem-solving, and workplace discipline. I also gained a better understanding of the challenges that may arise during the planning and implementation of municipal infrastructure projects.

This chapter presents the knowledge, skills, and professional experience gained during the internship. It also discusses the challenges faced, the solutions and lessons learned, and the limitations encountered during the training period.

3.1 Roles and Responsibilities

During the three-month civil engineering internship at the Infrastructure Department of Tarakeshwor Municipality, my roles and responsibilities involved both office-based engineering activities and field-based construction observation. The internship provided practical exposure to municipal engineering procedures, estimation, rate analysis, surveying, construction supervision, and documentation. My major roles and responsibilities were as follows:

- **Reviewing Municipal Documents and Procedures:** Studied and reviewed municipal construction-related documents and procedures during the initial weeks of the internship. This included project proposals, Bills of Quantities (BOQs), estimates, valuation documents, municipal construction approval procedures, map renewal (Naksa Navikaran), and property ownership transfer (Gharjagga Naamsaari).
- **Site Inspection and Construction Observation:** Participated in visits to several ongoing municipal projects, including the Old Age Home in Ward No. 8, the gabion wall project in Ward No. 6, and newly constructed intermediate-lane flexible roads in Wards No. 7 and 8. Construction activities, site conditions, materials, workmanship, and technical practices were observed under the guidance of municipal technical staff.
- **Site Measurement and Boundary Verification:** Assisted surveyors and technical staff in verifying roadside boundaries at the Old Age Home site. Measurement

tapes were used to take field measurements, which were recorded in a field book. A cadastral map (Naapi Naksa) with a scale of 1:500 was used for boundary verification. Points were marked using pegs and chalk on rocks or existing structures, and the measurements were discussed and verified with the concerned landowners.

- Observation of RCC Construction: Observed the reinforcement laying and formwork of the reinforced-concrete retaining compound wall at the Old Age Home site at different stages of construction. Reinforcement dimensions and construction details were also checked with the supervisor.
- Observation of Gabion Wall Construction: Observed the construction of a stepped gabion wall along a steep roadside embankment in Ward No. 6. The work included stepped foundation and wall construction, assembly of mechanically woven double-twisted heavy-zinc-coated hexagonal gabion boxes, installation of diaphragms, lacing and bracing wires, and hand placement of hard and durable angular stones. Dimensions of the work were checked with the supervisor.
- Observation of Road-Marking Works: Observed thermoplastic road marking on newly constructed narrow two-lane intermediate-lane flexible roads in Wards No. 7 and 8. The activities included surface sweeping, string alignment, machine application of thermoplastic material, dimensional checking, yellow edge-line marking, and black-and-white zebra crossing installation. Attention was given to surface cleanliness and dryness before marking.
- Preparation of Rate Analysis: Prepared rate analyses for several proposed building and masonry works under the supervision of Assistant Engineer Rakesh S. Magar. Material, labour, and equipment data were entered into an existing municipal Excel template. Coefficients were obtained from DUDBC norms and the Department of Roads norms, while Kathmandu District approved rates for FY 2081/82 were used for applicable items.
- Cost Calculation and Rate Analysis: Calculated rates generally on a 1 cubic metre basis, where applicable, by considering materials, skilled, semi-skilled and unskilled labour, and equipment. The material, labour, and equipment costs were summed and the applicable contractor overhead of 15%, consisting of 10% profit and 5% overhead, was added as required by the rate-analysis procedure.
- Preparation of Truss Structure Estimate: Independently prepared a complete estimate for a proposed truss structure above an existing one-storey stone-masonry building within the municipality office premises. The proposed structure was intended for municipal document storage. The existing building was measured, and quantities were calculated for the proposed floor/slab, lightweight sandwich-panel walls, doors, windows, staircase, vertical supports, base plates, and other required components.
- Comparison of Structural Alternatives: Prepared estimates for two alternatives for the proposed truss structure: an RCC slab floor and a mild-steel plate floor. Both alternatives retained and reinstalled the existing mild-steel truss as the roof. The estimates were prepared in a single Excel sheet and submitted to the supervisor

for comparison. Based on the initial cost comparison, the RCC slab alternative was found to be less costly.

- Use of Engineering Software and Reference Documents: Used the METWET application to determine unit weights of different metal sections and plates required for the truss estimation. Relevant DUDBC norms, Department of Roads norms, and approved district rate documents were consulted to obtain appropriate coefficients and rates.
- Documentation and Coordination: Completed engineering worksheets and rate-analysis sheets were printed and filed after review. Technical discussions with the supervisor and other municipal staff helped clarify estimation procedures, construction practices, and the advantages and limitations of different engineering alternatives.

Through these roles and responsibilities, the internship provided practical experience in connecting academic knowledge with actual municipal engineering practices. In particular, the knowledge gained from the Estimation and Valuation course during the fifth semester was applied directly while preparing rate analyses and project estimates.

3.2 Weekly Log

The following table summarises the technical activities performed each week, based on the daily work-progress record maintained throughout the internship.

Week	Technical Activities	Tools/Software
1	Orientation and Familiarization	Computer, MS Office, municipal document files and records
2	Municipal Documents and Procedures	Computer, MS Office, municipal document files, cadastral/map records
3	Site Visit and Boundary Verification	Measurement tape, field book, 1:500 cadastral map (Naapi Naksa), pegs, chalk
4	RCC Construction Observation and Site Activities	Measurement tape, reinforcement measuring tools, engineering drawings, field book
5	Rate Analysis	Microsoft Excel , DUDBC norms, Department of Roads rate-analysis norms, Kathmandu District approved rate schedule 2081/82
6	Rate Analysis and Cost Calculation	Microsoft Excel , DUDBC norms, Department of Roads norms, Kathmandu District approved rate schedule 2081/82

7	Gabion Wall Construction	Measurement tape, measuring scale/tape, engineering drawings/site documents, gabion boxes
8	Preparation of Truss Structure Estimate	Microsoft Excel, METWET , calculator, measuring tape, DUDBC norms, Kathmandu District approved rate schedule
9	Truss Estimate and Alternative Comparison	Microsoft Excel, METWET, calculator, DUDBC norms, Kathmandu District approved rate schedule
10	Road and Thermoplastic Marking Observation	Thermoplastic road-marking machine, measuring tape, string/alignment tools, broom/sweeping tools
11	Review of Engineering Activities	Microsoft Excel, engineering drawings, municipal documents, calculator, measurement tools
12	Final Review and Documentation	Microsoft Excel, MS Office, municipal documents, printer, calculator

3.3 Description of the Project(s) Involved During Internship

During the three-month internship at the Infrastructure Department of Tarakeshwor Municipality, I was exposed to several ongoing and proposed municipal projects through field visits, site measurements, construction observations, estimation, and rate-analysis activities. The major projects and works involved during the internship included the Old Age Home construction project in Ward No. 8, gabion wall construction in Ward No. 6, intermediate-lane flexible road projects in Wards No. 7 and 8, and a proposed truss structure within the municipality office premises.

3.3.1 Old Age Home Construction Project – Ward No. 8

The Old Age Home construction project was one of the major sites visited during the internship. The site was located on sloping ground in Ward No. 8 of Tarakeshwor Municipality. The field visits provided practical exposure to site-boundary verification, surveying, measurement, setting out, and reinforced-concrete construction activities.

During the site visit, I assisted municipal surveyors and technical staff in verifying the roadside boundary of the site in the presence of engineers and other municipal personnel. The purpose of the measurement was to verify the actual boundary of the site against the available land records. A cadastral map (Naapi Naksa) at a scale of 1:500 was used as a reference for the boundary verification.

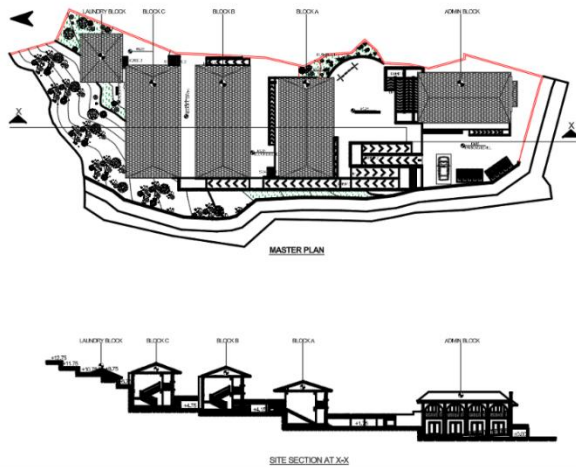


Figure 4 Old Age House Masterplan



Figure 3 Google Map Image of Old Age House site

The measurements were taken using measurement tapes and recorded in a field book. Points identified during the boundary verification were marked using pegs and chalk on suitable locations such as rocks and existing structures. The measurements and boundary information were also discussed with the concerned landowners, who agreed with the land information provided during the verification process. This activity helped in understanding the importance of accurate surveying and land-boundary verification before and during municipal construction works.



Figure 5 Formwork Removal



Figure 6 Concreting Work

During subsequent visits, I observed the construction of a reinforced-concrete retaining compound wall around the site. Reinforcement laying and formwork were observed at different stages of construction. Under the guidance of the supervisor, I checked reinforcement dimensions and observed the arrangement of reinforcement and formwork before concrete placement. The site visits provided practical understanding of the relationship between site conditions, drawings, measurements, reinforcement detailing, and construction execution.

3.3.2 Gabion Wall Construction Project – Ward No. 6

Another important project observed during the internship was an ongoing gabion wall construction project in Ward No. 6. The project was associated with planned road development and was located along the uphill side of a steep roadside embankment.



Figure 8 Gabion Wall



Figure 7 Road Erosion

The existing roadside condition was vulnerable to erosion because rainfall and direct surface runoff were washing away soil from the road edge. The steep embankment also created a potential risk of slope instability and landslide-related damage. Therefore, the gabion wall was being constructed to protect the roadside, control soil erosion, and improve the stability of the embankment.

The construction involved a stepped foundation and stepped gabion wall arrangement suitable for the sloping terrain. The gabion boxes observed at the site were mechanically woven, double-twisted, heavy-zinc-coated hexagonal wire-mesh units. Internal diaphragms were provided within the baskets to maintain their shape and improve structural stability. Lacing and bracing wires were used to connect and strengthen the individual gabion units.

Hard and durable angular stones obtained from quarry or natural sources were placed manually inside the gabion boxes. Careful placement of the stones was observed to ensure proper filling and interlocking within the baskets. The dimensions and construction details were checked with the supervisor during the site visit.

Since the road remained open to traffic during construction, coloured plastic barriers were used to provide basic separation between the construction area and road users. The project provided practical exposure to slope protection, erosion control, gabion

construction techniques, site safety, and the application of retaining structures in road projects.

3.3.3 Intermediate-Lane Flexible Road Marking Project – Wards No. 7 and 8

Field visits were also made to newly constructed intermediate-lane flexible roads in Wards No. 7 and 8. These roads were relatively narrow two-lane roads constructed using flexible pavement. The visits provided exposure to road finishing activities, particularly the application of thermoplastic road markings.



Figure 10 Road Marking

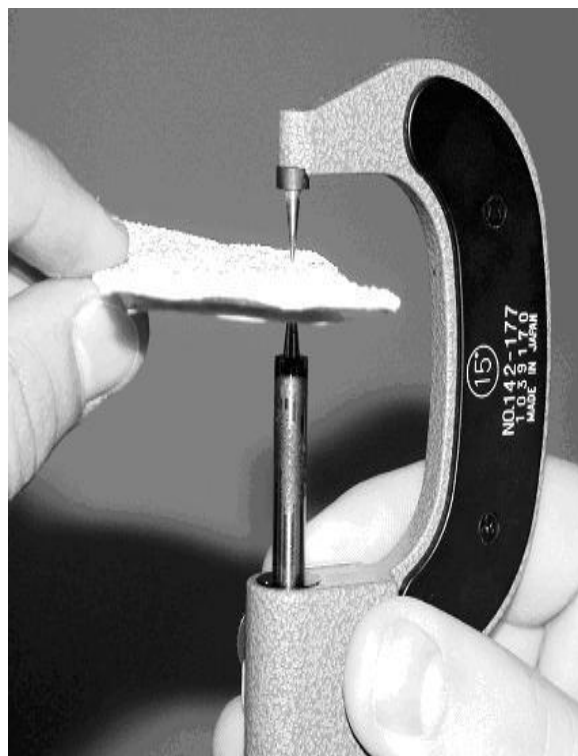


Figure 9 Needlepoint Micrometre

During the road-marking work, the pavement surface was first swept and cleaned to remove dust, loose particles, and other contaminants. String alignment was then used to establish the required position of the markings. A road-marking machine was used to apply the thermoplastic material to the prepared pavement surface.

The dimensions and alignment of the markings were checked during the work. Yellow edge lines were observed along the sides of the road, while black-and-white zebra crossings were provided at designated locations. Thickness of the road marking was checked using needlepoint micrometer. The condition of the pavement surface had a direct effect on the quality of the marking. Dust accumulation and wet pavement were observed to interfere with the marking process; therefore, the surface was cleaned and the thermoplastic marking was applied only when the pavement was sufficiently dry.

No dedicated traffic-control arrangement was observed during the road-marking activity. The observation provided practical understanding of road-marking procedures, surface

preparation, alignment, dimensional control, and the importance of suitable weather and pavement conditions for thermoplastic application.

3.3.4 Estimation of a Truss Structure for Municipal Document Storage

An important office-based project involved the preparation of a complete estimate for a proposed truss structure above an existing one-story stone-masonry building within the perimeter of the municipality office. The proposed structure was intended solely for storing municipal documents in a library-like facility.

The floor area of the proposed structure was 12.5 m × 5.5 m. The existing one-story stone-masonry building was measured to determine the existing dimensions and to prepare the proposed estimate. The existing mild-steel truss roof was planned to be removed and reinstalled above the new upper structure as the roof in both alternatives. The truss was to be supported by vertical members connected to base plates.

Two structural alternatives were assigned for comparison:

The first alternative involved removing the existing truss and constructing an RCC slab floor. The proposed slab was a 4-inch thick M20 one-way slab reinforced with 10 mm diameter reinforcement in both longitudinal and lateral directions at 4-inch c/c. Formwork was also included in the estimate. After construction of the upper floor, the existing mild-steel truss was to be reinstalled above it as the roof.

The second alternative retained the concept of the truss roof but provided a 6 mm thick mild-steel plate floor over the same floor area. The estimate included MS angular sections, MS hollow pipes, welding, bolted base plates, and primer. The existing truss was similarly intended to be reinstalled above the new structure.

Both alternatives included the construction of external walls using 75 mm EPS and cement-based lightweight sandwich panels, with 4.5 mm non-asbestos calcium-silicate face boards on both sides. The estimates also included doors, windows, a metal staircase, vertical supports, base plates, and other necessary construction components. Door and window openings were deducted from the wall quantities during quantity calculation. Masonry work was included where required.

The estimate was prepared using applicable Kathmandu District approved rates and DUDBC norms. Labor requirements were included, and the applicable district rates incorporated contractor profit and overhead as required. METWET was used to determine the weight per meter and weight per square meter of the metal plates and other metal sections required for the project.

Both alternatives were prepared in a single Excel sheet and compared based on their initial construction cost. The estimation showed that the RCC slab alternative was less costly than the mild-steel plate alternative. The completed estimate and comparison were independently prepared and submitted to the Assistant Engineer for checking. The supervisor also discussed the advantages and disadvantages of both alternatives,

providing practical insight into structural-material selection, cost comparison, and engineering decision-making.

3.3.5 Rate Analysis of Proposed Building and Masonry Works

In addition to the major field projects, I was involved in preparing rate analyses for several proposed building and masonry works during multiple office sessions. The work was carried out under the supervision of Assistant Engineer Rakesh S. Magar using an existing municipal Excel template.

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	ख) ज्यामी	३०	संख्या	९३५	२८०५०		
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	ग) बालुवा	४.२	घ.मी	३१७८.३२	१३३४८.९४४		
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Figure 11 Sample of Rate Analysis Work

Material, labor, and equipment coefficients were obtained from the applicable DUDBC norms and the Department of Roads Norms for Rate Analysis of Road and Bridge Works. For applicable items, Kathmandu District approved rates for fiscal year 2081/82 were used because Tarakeshwor Municipality lies within Kathmandu District. Market rates were considered only for items that were not available in the district-approved rate schedule.

The analysis generally considered a 1 cubic meter quantity basis, where applicable. The items considered included cement, sand, aggregate, bricks, reinforcement steel, stone, wire mesh, timber, paint, bitumen, and other construction materials. Labor categories such as skilled, semi-skilled, and unskilled labor were included according to the nature of the work. Equipment costs were also included where required.

The total material, labor, and equipment costs were calculated and summed. A 15% contractor overhead, consisting of 10% profit and 5% overhead, was then added to obtain the final analyzed rate where applicable. The completed rate-analysis sheets were checked by the supervisor before being printed and filed. The work provided direct practical application of the concepts learned in the fifth-semester Estimation and Valuation course.

During the analysis, it was also observed that market rates for several materials were generally higher than the corresponding approved district rates. This highlighted the importance of selecting the appropriate rate source and understanding the difference between approved schedule rates and prevailing market prices during project estimation.

3.3.6 Overall Project Exposure

The projects and activities observed during the internship covered several important areas of municipal civil engineering, including **surveying and land-boundary verification, building construction, RCC works, slope protection, gabion structures, flexible road construction, thermoplastic road marking, quantity estimation, rate analysis, and cost-based comparison of engineering alternatives.**

The combination of field and office activities provided an opportunity to understand the complete relationship between site conditions, technical documents, measurements, quantities, rates, construction methods, and project costs. The experience also demonstrated the role of municipal technical personnel in planning, estimating, supervising, documenting, and implementing infrastructure projects.

3.4 Tasks / Activities Performed

During the three-month internship at the Infrastructure Department of Tarakeshwor Municipality, I performed and participated in the following major tasks and activities:

1. Study of Municipal Documents and Procedures

- Reviewed project proposals, Bills of Quantities (BOQs), estimates, valuation documents, engineering drawings, and other municipal construction records.
- Studied procedures related to **Naksa Navikaran (map renewal), Gharjagga Naamsaari (property ownership transfer)**, and municipal construction approval.
- Observed the general workflow of municipal projects from proposal preparation and estimation to implementation, measurement, valuation, and documentation.

2. Site Measurement and Boundary Verification

- Assisted surveyors and technical staff in verifying the roadside boundary of the Old Age Home site in **Ward No. 8.**
- Used **measurement tapes** to take field measurements and recorded the measurements in a field book.
- Used a **1:500 cadastral map (Naapi Naksa)** to compare mapped land information with actual site conditions.
- Helped mark verified points using **pegs and chalk** on rocks and existing structures.
- Communicated with concerned landowners during the boundary-verification process.

3. Observation of RCC Retaining Compound Wall

- Observed reinforcement laying and formwork activities at the Old Age Home construction site.
- Checked reinforcement dimensions and arrangements under the guidance of the supervisor.

- Observed construction activities at different stages and related the site work with engineering drawings and technical requirements.
4. **Preparation of Rate Analysis**
- Prepared rate analyses for proposed building and masonry works using the municipality's **Microsoft Excel template**.
 - Entered material, labour, and equipment coefficients and rates into the spreadsheet.
 - Used **DUDBC norms** and the **Department of Roads Norms for Rate Analysis of Road and Bridge Works** for applicable coefficients.
 - Used **Kathmandu District approved rates for FY 2081/82** for applicable construction items.
 - Used market rates only for items not available in the approved district schedule.
 - Considered cement, sand, aggregate, bricks, reinforcement steel, stone, wire mesh, timber, paint, bitumen, and other materials.
 - Included skilled, semi-skilled, and unskilled labour and equipment where required.
 - Prepared analyses generally on a **1 m³ basis**, where applicable.
 - Calculated material, labour, and equipment costs and added **15% contractor overhead**, comprising 10% profit and 5% overhead, where applicable.
 - Submitted the completed rate analyses to the supervisor for checking before printing and filing.
5. **Preparation of Truss Structure Estimate**
- Independently prepared a complete estimate for a proposed truss structure above an existing one-storey stone-masonry building within the municipality office premises.
 - Measured the existing building and used the floor dimensions of **12.5 m × 5.5 m** for the proposed structure.
 - Prepared estimates for two alternatives:
 - **RCC slab floor alternative:** 4-inch M20 one-way slab with 10 mm reinforcement in longitudinal and lateral directions at 4-inch c/c, including formwork.
 - **Mild-steel floor alternative:** 6 mm mild-steel plate with MS angular sections, MS hollow pipes, welding, bolted base plates, and primer.
 - Included quantities for the floor/slab, lightweight sandwich-panel walls, doors, windows, metal staircase, vertical supports, base plates, and other required components.
 - Considered **75 mm EPS and cement-based lightweight sandwich panels** with 4.5 mm non-asbestos calcium-silicate face boards on both sides.
 - Deducted door and window openings from the calculated wall quantities.
6. **Use of METWET for Metal Weight Calculation**
- Used the **METWET application** during the truss estimation process.
 - Determined the **weight per metre of metal sections** and **weight per square metre of metal plates**.

- Used the obtained unit weights to calculate the total weight and cost of required mild-steel components.
7. **Comparison of Structural Alternatives**
- Prepared both truss-floor alternatives in a single Excel worksheet.
 - Compared the estimated initial construction costs of the RCC slab and mild-steel plate alternatives.
 - Found that the **RCC slab alternative was less costly** based on the initial cost comparison.
 - Submitted the completed estimate and comparison to the supervisor for review.
 - Discussed the advantages and disadvantages of both alternatives with the supervisor.
8. **Observation of Gabion Wall Construction**
- Observed the construction of a gabion wall along the uphill side of a steep roadside embankment in **Ward No. 6**.
 - Observed the construction of **stepped foundations and stepped gabion walls**.
 - Studied the assembly of mechanically woven, double-twisted, heavy-zinc-coated hexagonal wire-mesh gabion boxes.
 - Observed the installation of **internal diaphragms, lacing wires, and bracing wires**.
 - Observed the hand placement of hard and durable angular quarry or natural-source stones inside the gabion boxes.
 - Checked construction dimensions with the supervisor.
 - Observed the use of coloured plastic barriers while maintaining traffic movement during construction.
9. **Observation of Thermoplastic Road Marking**
- Observed thermoplastic road marking on newly constructed narrow two-lane intermediate-lane flexible roads in **Wards No. 7 and 8**.
 - Observed pavement sweeping and cleaning before marking.
 - Observed the use of **string alignment** to establish the position of road markings.
 - Observed thermoplastic application using a **road-marking machine**.
 - Observed the application of **yellow edge lines** and **black-and-white zebra crossings**.
 - Checked the dimensions and alignment of road markings during the work.
10. **Measurement of Road-Marking Thickness**
- Observed and participated in checking the thickness of thermoplastic road markings using a **needlepoint micrometer**.
 - The pointed measuring tip was used to take thickness measurements of the thermoplastic marking while minimizing interference from the glass beads on the marking surface.
 - Observed the importance of proper surface preparation and measurement for ensuring the required quality and thickness of road markings.

- Noted that dust and moisture affected the marking process; therefore, the pavement was cleaned and marking was carried out only when the surface was dry.

11. Field Observation and Construction Supervision

- Participated in repeated visits to ongoing municipal projects to observe construction progress.
- Observed construction methods, material use, workmanship, dimensions, and site conditions.
- Discussed observed technical details with engineers, supervisors, and other municipal technical staff.
- Related practical construction activities to concepts learned during academic study.

12. Documentation and Technical Record Keeping

- Completed estimation and rate-analysis sheets were reviewed by the supervisor before being printed and filed.
- Reviewed and organized technical documents and records related to municipal projects.
- Participated in technical discussions regarding construction methods, estimation, material selection, and project costs.

CHAPTER 4: CONCLUSION AND LEARNING OUTCOMES

This chapter presents the overall conclusion of the internship and provides recommendations based on the knowledge, skills, experiences, and limitations encountered during the training.

4.1 Conclusion

The three-month internship at the Infrastructure Department of Tarakeshwor Municipality, completed from 23 April 2026 to 23 July 2026, was a valuable and practical part of my BE Civil program at Nepal College of Information Technology. It provided me with an opportunity to connect academic knowledge with real civil engineering activities carried out in a municipal working environment.

During the internship, I gained exposure to municipal documentation, cadastral map interpretation, site measurement, boundary verification, site marking, construction observation, rate analysis, project estimation, and cost comparison. I also observed different infrastructure works, including the construction of an Old Age Home, a gabion wall for roadside protection, and road-marking activities on newly constructed flexible roads.

The preparation of rate analyses using DUDBC norms, Department of Roads norms, Kathmandu District approved rates, and Microsoft Excel improved my understanding of construction costing and resource requirements. Similarly, preparing estimates for RCC slab and mild-steel plate alternatives for a truss-based document-storage structure helped me understand how engineers compare construction options and make economical decisions. The use of METWET also introduced me to the practical calculation of steel unit weights.

The internship helped me develop both technical and professional abilities. I improved my skills in field observation, document handling, measurement recording, Excel-based estimation, technical communication, teamwork, coordination, time management, and professional responsibility. I also learned the importance of accuracy, proper documentation, supervision, safety awareness, and verification in civil engineering work.

Overall, the internship provided a meaningful understanding of the responsibilities of civil engineers in municipal infrastructure development. It showed me that civil engineering involves not only theoretical calculations but also field judgment, coordination, cost evaluation, public responsibility, and practical problem-solving. The knowledge and experience gained during the training have strengthened my confidence and will support my future academic work and professional career as a civil engineer.

4.2 Learning Outcomes

The internship provided learning opportunities in both technical and professional aspects of civil engineering. By participating in office activities and field visits, I was able to relate academic concepts to practical engineering applications. The major learning outcomes are presented below.

4.2.1 Skills Acquired

- **Surveying and Measurement Skills:** Gained practical experience in using measurement tapes, recording measurements in a field book, identifying site points, and assisting in boundary verification.
- **Cadastral Map Interpretation:** Developed an understanding of how **1:500 cadastral maps (Naapi Naksa)** can be used with field measurements to verify land boundaries and site information.
- **Site Setting-Out Skills:** Learned basic methods of transferring identified points and boundaries onto the actual site using pegs, chalk, and other marking methods.
- **Quantity Estimation Skills:** Improved the ability to calculate quantities of construction materials and work items from available dimensions, drawings, and site measurements.
- **Rate Analysis Skills:** Developed practical skills in preparing rate analyses by considering material, labour, and equipment components and applying appropriate coefficients and approved rates.
- **Use of Estimation Norms and Rate Schedules:** Gained experience in referring to **DUDBC norms, Department of Roads norms, and Kathmandu District approved rates for FY 2081/82** to obtain appropriate coefficients and rates.
- **Excel-Based Engineering Calculations:** Improved proficiency in **Microsoft Excel** for quantity calculations, rate analysis, cost estimation, and comparison of project alternatives.
- **Metal Weight Calculation:** Learned to use the **METWET application** to obtain unit weights of metal sections and plates for estimation purposes.
- **RCC Construction Knowledge:** Developed practical understanding of reinforcement laying, reinforcement dimension checking, and formwork through observation of the RCC retaining compound wall.
- **Gabion Construction Knowledge:** Learned about gabion box assembly, diaphragms, lacing and bracing wires, stone filling, stepped construction, and the application of gabion walls for erosion control and slope stabilization.
- **Road Construction and Marking Knowledge:** Gained practical understanding of intermediate-lane flexible roads and thermoplastic road marking, including surface preparation, alignment, machine application, edge-line marking, and zebra-crossing marking.
- **Road-Marking Quality Checking:** Gained exposure to checking thermoplastic road-marking thickness using a **needlepoint micrometer** and understanding the importance of clean and dry pavement surfaces during marking.
- **Cost Comparison and Decision-Making:** Developed an understanding of comparing alternative construction solutions based on estimated initial costs, as

demonstrated through the RCC slab and mild-steel plate alternatives for the proposed truss structure.

- **Technical Documentation:** Developed skills in preparing, reviewing, printing, organizing, and filing engineering documents such as estimates, rate analyses, BOQs, and related technical records.
- **Construction Observation Skills:** Improved the ability to observe construction activities systematically and identify important aspects such as dimensions, materials, workmanship, construction methods, and site conditions.
- **Communication and Coordination:** Improved communication skills through interaction with engineers, surveyors, technical staff, landowners, and other stakeholders during field and office activities.
- **Professional Skills:** Developed greater awareness of engineering ethics, responsibility, teamwork, technical supervision, time management, and the importance of accuracy in civil engineering work.
- **Application of Academic Knowledge:** Strengthened the ability to apply theoretical concepts, particularly those related to **Estimation and Valuation**, to practical engineering tasks carried out in a municipal organization.

4.2.2 Recommendations

Based on my internship experience at the Infrastructure Department of Tarakeshwor Municipality, the following recommendations are proposed:

- Internship students should receive a clear orientation about the municipality, department responsibilities, project procedures, and workplace expectations before the beginning of their training.
- Interns should be given opportunities to use engineering tools and software such as Microsoft Excel, AutoCAD, Advanced features in Google Earth, KML files, and METWET.
- Greater exposure should be provided to surveying instruments, construction-material testing, structural inspection, and other practical engineering equipment.
- Field visits should be conducted with proper safety precautions, including suitable protective equipment, work-area barriers, and clear instructions for interns.
- Technical personnel should maintain proper records of site visits, measurements, construction progress, material quality, and project observations.
- Coordination among municipal departments, ward offices, contractors, consultants, users' committees, and local communities should be strengthened for efficient project implementation.
- Current market rates should be periodically reviewed for items that are not included in the approved district-rate schedule.

- Municipalities should continue using suitable erosion-control, slope-protection, drainage, and road-safety measures according to site conditions.
- Colleges should maintain regular communication with host organizations and monitor the progress and learning outcomes of their students.
- Students should be encouraged to maintain field notes, record their activities regularly, and collect relevant information throughout the internship.
- Future interns should take initiative, ask questions, follow safety instructions, accept feedback, and maintain professional conduct during their training.

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